# Symmetrix-Polar

## Consolidated speed and accuracy comparison

MACE-Polar (PolarMACE) on the Symmetrix-XL nonlinear Kokkos engine

LAMMPS pair style, single A100, f64

Version: 2026-10-19

Model: polar_symmetrix_probe.json (MACE_Nonlinear + polar)

Repositories: Yong-Q/symmetrix-polar · starlessinred-interns/qiuyong

This document is generated from docs/COMPARISON_REPORT_2026-10-19.md. All numbers come from the repository JSON/log artifacts; in-flight results are marked pending. The force gate (1e-4 eV/Å) is treated as a hard minimum: RBE and 1-hop ghost MPI are rejected as production alternatives.

# Symmetrix-Polar: Consolidated Speed and Accuracy Comparison

**Date:** 2026-10-19 (v29 / v30 evidence cut-off, ~08:20 UTC; later measured rows are marked with their source logs) **Scope:** PolarMACE (MACE-Polar) support in the Symmetrix-XL C++/Kokkos engine. **Basis:** completed JSONs, gate logs, and reports only. No engine/model code was modified while preparing this document.

**Status legend: PASS** = measured and accepted against its gate; **FAIL** = measured and rejected; **pending** = work is in flight and no completed artifact exists yet. No pending result is included as a performance claim.

## 0. Summary

| Item | Result |
|---|---|
| Single-rank v29 speed, 128 atoms, deterministic | 183.78 ms/eval (v28: 6966.79 ms, 37.91x) |
| Single-rank v29 speed, 1024 atoms, deterministic | 1007.72 ms/eval (v28: 7795.59 ms, 7.74x) |
| Same-model ablation, 128 atoms | backbone-only 87.24 ms -> full PolarMACE 183.50 ms; polar layer adds 96.26 ms (2.10x) |
| Accuracy gates, modules 5-8 | all PASS |
| End-to-end engine vs PyTorch, 32 frames | 32/32 PASS, max energy error 1.034e-6 eV, max force error 1.624e-7 eV/A |
| LAMMPS single point, 1 rank / exact 2 rank | PASS / PASS (dE 2.91e-11 eV; max dF 2.62e-14 / 1.69e-6 eV/A) |
| LAMMPS NVT MD, 50 steps | PASS, no lost atoms, Etot span 0.21295 eV |
| RBE production use | rejected by the force gate (RMS_F 4.10e-2 to 1.03e-2 eV/A vs 1e-4 eV/A gate) |
| 1-hop MPI mode | rejected by the force/energy gates (max dF 4.116e-3 eV/A, dE 7.94e-5 eV) |
| MPI speed on one shared A100 | no speedup; exact 2-hop is 28.1x slower than 1 rank |
| MH-1/direct/JIT fast path | architecturally blocked; the upstream linear engine cannot load MACE_Nonlinear |
| PyTorch MACE-Polar reference speed | 94.0 ms/eval median (92.1 min) vs engine 183.8 ms -> engine 1.95x slower for the full model |

The v29 optimization removed two concrete bottlenecks: per-node-serial UVU tensor products and a ~0.7 s/eval device-environment probe on non-JIT models. All numerical gates are unchanged after the optimization.

## 1. Scope and model

### 1.1 What is being compared

All "same-model" statements below compare the same PolarMACE checkpoint (`MACE-POLAR-1-M.model`) exported to the engine JSON `polar_symmetrix_probe.json`, evaluated in the same 128-atom NH3 box (14.736126 A cubic, periodic) with the same float64 energy+force path. The linear MACE-OFF23 comparison is explicitly **not** the same model and is labelled wherever it appears.

| Property | Full PolarMACE | Nonlinear backbone (polar off) | Linear MACE-OFF23 medium |
|---|---|---|---|
| JSON model_type | MACE_Nonlinear | MACE_Nonlinear | MACE |
| Engine | MaceNonlinearKokkosT | MaceNonlinearKokkosT | MACEKokkos |
| JSON size (bytes) | 316,958,257 | 269,988,335 | 37,330,647 |
| Elements | 83 | 83 | 10 |
| r_cut | 6.0 A | 6.0 A | 5.0 A |
| Interactions / l_max | 2 / 3 | 2 / 3 | standard MACE |
| Channel pattern | 512 -> 128, nonlinear bias readout | 512 -> 128, nonlinear bias readout | linear readout |

| Property | Full PolarMACE | Nonlinear backbone (polar off) | Linear MACE-OFF23 medium |
|---|---|---|---|
| Polar layer | full SCF + Ewald/RBE | disabled | inert placeholder |
| Execution mode | materialized (no direct/JIT) | materialized | materialized or generic streamed |
| Typical use here | production target | same-model backbone ablation | engine-path reference only |

**Hardware/software:** one NVIDIA A100-80GB per run, CUDA 12 sm80 backend, Kokkos 5.x, float64, energy+forces. Times are LAMMPS `Loop time` or Python `median_ms` over 20 evals, as noted. The GPU was shared during part of the campaign; the v29 final figures are the uncontended reruns. Absolute times are therefore contention-sensitive, while same-run ratios are stable.

## 1.2 Linear vs nonlinear engines

"Linear" and "nonlinear" here are Symmetrix engine/model families, not linear-vs-nonlinear MLIPs:

| | MACEKokkos (linear engine) | MaceNonlinearKokkosT (nonlinear engine) |
|---|---|---|
| Accepted model_type | MACE, MACEField | MACE_Nonlinear |
| JSON format | format-version 2 compact radial | format-version 3 nonlinear |
| Readout | standard linear MACE readout | nonlinear readout (nonlinear, NonLinearBiasReadoutBlock for PolarMACE) |
| Polar blocks | polar_blocks_enabled() returns false | full polar block set and SCF pipeline |
| Direct gate | supports_streamed_edges() = uses_compact_radial | supports_streamed_edges() = mh1_fast_path |
| H1 MPI | first-interaction hidden node state | materialized node-feature exchange only |

**The upstream linear Kokkos engine cannot load MACE_Nonlinear at all.** `MACEKokkos::load_from_json` rejects it at `mace_kokkos_model.cpp:165-167` with:

```
MACE_Nonlinear JSON must be loaded through the nonlinear MACE Kokkos evaluator,
not legacy MACEKokkos.
```

The naming "H1" is a collision: in the linear engine H1 is the first-interaction hidden state; it is not the MACE-MH-1 model family. No PolarMACE layer can be delegated to `MACEKokkos`, and the linear engine's `mpi_message_passing` path cannot be reused for PolarMACE.

## 1.3 MH-1 fast-path status

PolarMACE cannot use the current MACE-MH-1 direct/JIT fast path. The engine's family descriptor rejects it with:

```
MH-1 node, edge, and hidden irreps are inconsistent.
```

This is architectural, not an export defect. The original checkpoint has the same architecture as the JSON. The four deviations from the admitted MH-1 family are:

1. second interaction uses `128x0e+128x1o` edge features instead of the first interaction's `512x0e+512x1o`;
2. second interaction keeps a vector component in `hidden_irreps` (`512x0e+512x1o` instead of scalar-only `512x0e`);
3. second product returns `512x0e+512x1o` instead of scalar-only `512x0e`;
4. second readout is `NonLinearBiasReadoutBlock` (with a learned `linear_mid` stage) instead of `NonLinearReadoutBlock`.

A v3 `MH1_UVU` contract is present in the JSON, but it covers only the per-interaction UVU tensor products and does not establish v4 node-program family compatibility. The v4 contract generator fails first at `products[1] output irreps are outside the MH-1 support matrix`. Re-exporting (Route A) cannot fix this without changing the learned function; relaxing only the gate (Route C) is numerically incorrect and is also not a speed win. See `docs/design/MH1_FASTPATH_FEASIBILITY.md` for the full rejection chain and the staged B1-B4 route assessment. No speedup from MH-1 is claimed anywhere in this report.

# 2. Accuracy and correctness gates

## 2.1 Module gates 5-8 and end-to-end

| Gate | Metric | Tolerance | Measured | Verdict |
|---|---|---|---|---|
| Module 5, density (12 items) | max relative density error | rel <= 1e-6 | 4.22e-8 | PASS |
| Module 5, density (12 items) | max absolute density error | abs <= 5e-6 | 6.689e-8 | PASS |
| Module 5, stages (11 items) | max relative stage error | rel <= 1e-6 | 7.13e-8 | PASS |
| Module 5, pairing | engine state -> fixture step mapping | exact | ['step1','step2','step2'] for all 4 molecules | PASS |
| Module 6, Coulomb | max rho-hat relative error | rel <= 1e-6 | 1.26e-15 | PASS |
| Module 6, Coulomb | max dE relative / absolute | rel <= 1e-6; abs <= 5e-6 eV | 8.43e-16 / 5.551e-16 eV | PASS |
| Module 6, Coulomb | max end-to-end energy error | informational | 1.491e-8 eV | PASS |
| Module 7, local electron | max stage absolute error | abs <= 5e-6 or rel <= 1e-6 | 2.230e-5 abs (passes on the relative criterion) | PASS |
| Module 7, local electron | max output absolute error | abs <= 1e-6 | 4.038e-7 | PASS |
| Module 8, forces | max finite-difference error | <= 1e-3 | 5.783e-4 | PASS |
| Module 8, forces | max parity vs PyTorch | <= 1e-3 | 7.191e-7 | PASS |
| Module 8, forces | max sum-of-forces residual | exact translation invariance | 1.776e-14 | PASS |
| End-to-end, 32 frames | frames within tolerance | 32/32 | 32/32 | PASS |
| End-to-end, energy | max relative / absolute | rel <= 1e-6; abs <= 5e-6 eV | 1.1e-11 / 1.034e-6 eV | PASS |
| End-to-end, forces | max force error | <= 1e-4 eV/A | 1.624e-7 eV/A (mean max 9.456e-8) | PASS |
| End-to-end, geometry | max position drift | 0 | 0.0 A | PASS |

Notes: - Module 5 evaluated 12 density items (4 molecules x 3 gated step comparisons) plus 11 stage tensors. The separate "anchor" field-independent density comparison (rel up to 9.57e-8) is recorded but not part of the module-5 gate. - Module 7 passes because its criterion is `abs <= 5e-6 OR rel <= 1e-6`; the worst absolute stage item (co2-f1, 2.230e-5) is at reference scale 351.146 and passes on the relative criterion. - Module 8 also reports `|polar|max` between 3.0e-3 and 4.7e-2 eV/A across configurations, confirming the polar force term is not negligible.

## 2.2 RBE statistical gate

Source: `results/polar_rbe_gate_v29.json`, log `polar_rbe_gate_v29opt.log`.

| Metric | Tolerance / criterion | Measured | Verdict |
|---|---|---|---|
| Slope of log RMS_F vs log p | ideal -0.5 | -0.49885 | PASS |
| Same-seed reproducibility | bitwise energy; dF ~ 0 | dE = 0; max dF = 6.141e-16 eV/A | PASS |
| Different-seed response | non-zero | dE = 0.1003 eV; dF = 9.365e-2 | PASS |
| RMS_F at p = 50 | decreasing | 4.0988e-2 eV/A | statistical PASS |
| RMS_F at p = 200 | decreasing | 2.1426e-2 eV/A | statistical PASS |
| RMS_F at p = 800 | decreasing | 1.0280e-2 eV/A | statistical PASS |
| Mean dE at p = 50 / 200 / 800 | unbiased (n=10) | -5.139e-2 / -2.344e-2 / +1.469e-3 eV | PASS |
| Max abs dE at p = 50 / 200 / 800 | monitored | 1.890e-1 / 1.426e-1 / 2.936e-2 eV | monitored |

The RBE estimator is unbiased and converges at the expected -1/sqrt(p) rate, but its force RMS at every measured p is **two to three orders of magnitude above the 1e-4 eV/A production force gate**. Under the project's gate-as-minimum rule (a speed mode is usable only if every accuracy gate still passes), RBE is accepted as a sampling/convergence gate but **rejected for production force evaluation**. See Section 3.6 for its 33 A timings.

## 2.3 LAMMPS pair-style and MD gates

Sources: `results/polar_lammps_parity.json`, `results/polar_lammps_parity_mpi2.json`,

`results/polar_md.json`, logs `polar_lammps_parity_v29final.log`, `polar_lammps_md_v29final.log`, `polar_lammps_md_mpi2_n32.log`.

| Gate | Metric | Tolerance | Measured | Verdict |
|---|---|---|---|---|
| LAMMPS single point, 1 rank | dE vs engine | abs <= 5e-5 eV | 2.910e-11 eV | PASS |
| LAMMPS single point, 1 rank | max dF vs engine | <= 1e-4 eV/A | 2.622e-14 eV/A | PASS |
| LAMMPS single point, 1 rank | max position drift | monitored | 4.974e-14 A | PASS |
| LAMMPS single point, exact 2 rank | dE vs engine | abs <= 5e-5 eV | 2.910e-11 eV | PASS |
| LAMMPS single point, exact 2 rank | max dF vs engine | <= 1e-4 eV/A | 1.687e-6 eV/A | PASS |
| LAMMPS single point, 1-hop 2 rank | dE vs engine | abs <= 5e-5 eV | 7.937e-5 eV | FAIL |
| LAMMPS single point, 1-hop 2 rank | max dF vs engine | <= 1e-4 eV/A | 4.116e-3 eV/A | FAIL |
| LAMMPS NVT MD, 128 atoms | completion / lost atoms / finite | 50/50, none, finite | 50/50, none, finite | PASS |
| LAMMPS NVT MD, 128 atoms | Etot span (relative to first step) | monitored | 0.21295 eV (25 fs) | PASS |
| LAMMPS NVT MD, 128 atoms | temperature | 300 K target | mean 276.81 K, min 188.96, max 335.95 | PASS |
| LAMMPS NVT MD, 128 atoms | pressure | monitored | 657.5 - 3231.3 bar | PASS |
| LAMMPS NVT MD, 128 atoms | throughput | monitored | 493.662 ms/step | PASS |
| MPI NVT MD, 32 atoms | completion / lost atoms / Etot span | 20 steps, none | 20/20, none, 0.013359 eV | PASS |
| MPI 4 rank, 128 atoms | runnable on one GPU | must fit device memory | ghost workspace 4.5-6.9 GiB; OOM | FAIL (resource) |

The 1-hop mode is 2.86x faster than exact 2-hop (1809.3 vs 5167.05 ms) but fails both the energy and force gates; it is not a usable approximation.

# 3. Speed

## 3.1 Single-rank v28 vs v29

Source: `mlip/bench/polar_perf_v28.json` and `results/polar_perf_v29.json` (20 evals each, median, energy+forces, 128-atom NH3 frame 0, 1024-atom 2x2x2 supercell).

| Configuration | v28 (ms/eval) | v29 (ms/eval) | Speedup | Delta |
|---|---|---|---|---|
| 128 atoms, deterministic | 6966.79 | 183.78 | 37.91x | -6783.0 ms |
| 128 atoms, RBE p = 50 | 6995.58 | 177.60 | 39.39x | -6818.0 ms |
| 128 atoms, RBE p = 200 | 6962.87 | 186.30 | 37.38x | -6776.6 ms |
| 128 atoms, RBE p = 800 | 7019.89 | 219.05 | 32.05x | -6800.8 ms |
| 1024 atoms, deterministic | 7795.59 | 1007.72 | 7.74x | -6787.9 ms |

Model load also dropped from ~48.0 s (v28) to ~12.4 s (v29) for the deterministic case (12.95 / 12.91 / 12.89 s for RBE p=50/200/800).

**Data-quality note.** The stage report quotes 6951.7 ms for the v28 deterministic row. That is the *energy-only* median in `perf_breakdown_v28.log`; the benchmark JSON for the same 20-eval energy+force run is 6966.79 ms. This report uses the JSON value so all v28/v29 rows use the same measurement definition.

## 3.2 Same-model ablation: backbone vs full PolarMACE

Source: `results/engine_compare.json` (LAMMPS single-step `Loop time`, 128-atom NH3 box, one A100).

| Configuration | Model | ms/eval | Relative to backbone |
|---|---|---|---|
| Nonlinear backbone only (polar blocks off) | polar_symmetrix_probe_polaroff.json | 87.24 | 1.00x |
| Full PolarMACE v29 | polar_symmetrix_probe.json | 183.50 | 2.10x |
| Polar layer increment (full - backbone) | same model, polar on vs off | 96.26 | 52.5% of full |
| leoff variant | polar_symmetrix_probe_leoff.json | pending | model exists (397,979,620 B); no timing JSON yet |

This is the clean same-model isolation: the nonlinear backbone is 87.2 ms and the polar layer adds 96.3 ms, i.e. the full model is 2.10x the backbone. The "leoff" variant (local-electron off) has a model file but no completed timing run, so it is marked pending rather than estimated.

## 3.3 Linear MACE-OFF23 materialized vs generic streamed

**Not the same model as PolarMACE.** This table isolates the *engine path* on one linear model (MACE-OFF23 medium: 37.33 MB, 10 elements, r_cut 5 A).

| Path | ms/eval | Notes |
|---|---|---|
| Materialized | 6.6102 | explicit streamed_edges materialized |
| Generic streamed | 19.2914 | requested auto, resolved generic |
| Auto -> generic (separate engine-compare run) | 7.219 | same resolution, 2.67x faster than the explicit generic run |

Observed run-to-run spread on the generic path is large at this small system (7.2 vs 19.3 ms). Materialized is consistently the fastest linear path here (6.61 ms), so "streamed vs materialized" is not the explanation for the linear-vs-PolarMACE gap. The correct reading of the old 7.2 ms vs 183.5 ms comparison is "linear 37 MB/r_cut 5 A model + MH-1-capable engine path" versus "270 MB backbone + nonlinear materialized path + polar layer"; it is **not** an apples-to-apples engine or accuracy comparison. Both engines are Kokkos/CUDA.

## 3.4 MPI: functional, but slower on one shared GPU

Sources: `results/polar_time_mpi1.json`, `polar_time_mpi2.json`, `polar_time_mpi2_onehop.json`, `linear_mpi.json`.

| Mode | 128-atom eval (ms) | Relative to 1 rank | Force gate | Verdict |
|---|---|---|---|---|
| 1 rank optimized | 183.59 | 1.00x | PASS | production path |
| 2 rank exact (2-hop ghost) | 5167.05 | 0.036x (28.1x slower) | max dF 1.687e-6 eV/A, PASS | functionally correct |
| 2 rank 1-hop ghost | 1809.30 | 0.102x (9.85x slower) | max dF 4.116e-3 eV/A, FAIL | rejected |
| Upstream linear MACE H1 message passing, 1 rank | 13.586 | 1.00x | n/a (linear model) | reference |
| Upstream linear MACE H1 message passing, 2 rank | 45.878 | 3.38x slower | n/a | reference |

The exact 2-hop nonlinear path is ~113x slower than the upstream linear H1 message-passing path (5167.05 vs 45.878 ms), but they are different models and different algorithms. On one shared A100, even the mature upstream linear MPI path is 3.38x slower at 2 ranks. True scaling requires a local graph with per-layer H1 feature exchange plus multiple GPUs.

## 3.5 Homogeneous 216-molecule per-step timings and extrapolations

Source: `mlip/bench/lammps_parity/homog_timing/homog_timing.json` (216 molecules, molecular density 0.006 A^-3, 0.5 fs timestep, 10-step LAMMPS runs, full PolarMACE v29). The N2 box is 33.019 A.

| Molecule | Atoms | Temperature (K) | ms/step | 10 ps | 100 ps | 1 ns |
|---|---|---|---|---|---|---|
| N2 | 432 | 151.43 (T/Tc=1.20) | 476.668 | 2.648 h | 26.482 h | 11.034 d |
| CO2 | 648 | 364.95 (T/Tc=1.20) | 1749.14 | 9.717 h | 97.174 h | 40.489 d |
| CH4 | 1080 | 317.00 (T*=2) | 1095.37 | 6.085 h | 60.854 h | 25.356 d |
| H2O | 648 | 776.52 (T/Tc=1.20) | 3649.87 | 20.277 h | 202.771 h | 84.488 d |
| NH3 | 864 | 486.48 (T/Tc=1.20) | 2217.59 | 12.320 h | 123.199 h | 51.333 d |

Extrapolations assume a 0.5 fs timestep and 100% duty cycle (10 ps = 20,000 steps; 100 ps = 200,000; 1 ns = 2,000,000). They are throughput estimates, not measured long runs. An earlier pilot log (`polar_homog_timing.log`) failed the N2 case; the completed JSON above is the valid run.

**Measured 1 ps N2 pilot.** `mlip/runs/n2-homog-pilot/run-status.json` records 500 warm-up + 2000 production steps (1.0 ps) at 0.5 fs: **537.99 ms/step**, no lost atoms, restart written. The final temperature is **185.48 K** against the 151.43 K target, so the 500-step warm-up is not sufficient for production sampling; a longer/stronger equilibration is required before extending this run to 0.1 ns or 1 ns. The per-step cost is consistent with the 10-step timing once trajectory dumps are included.

## 3.6 RBE on the 33 A N2 box

Source: `mlip/bench/lammps_parity/rbe_md_timing/rbe_md_timing.json` (432 atoms, N2 x216, 33.019 A cubic box, 10-step LAMMPS runs, per-step time).

| Mode | ms/step | Relative to deterministic |
|---|---|---|
| Deterministic (full k-space) | 477.827 | 1.000x |
| RBE p = 50 | 385.427 | 0.807x (-19.3%) |
| RBE p = 200 | 405.901 | 0.849x (-15.1%) |
| RBE p = 800 | 465.281 | 0.974x (-2.6%) |

**Caveat (gate-as-minimum rule).** These timings look attractive, especially p=50, but the same RBE approximation has force RMS errors of **4.099e-2 / 2.143e-2 / 1.028e-2 eV/A** at p=50/200/800 (Section 2.2), far above the **1e-4 eV/A** force gate. RBE therefore cannot be used for production force/MD work under the rule that a speed mode must pass every accuracy gate. A dedicated fixed-configuration force-error measurement on the 33 A box (`polar_rbe_force_error_33a.py`) is **in flight**; its JSON did not exist at the evidence cut-off, so no 33 A-specific force-error numbers are claimed here.

## 3.7 PyTorch MACE-Polar reference speed

Source: `mlip/logs/polar_torch_reference_speed.log` (measured 2026-10-19). Same 128-atom NH3 frame and the same `MACE-POLAR-1-M.model` checkpoint, `mace_polar`, float64, one A100; 2 warm-ups then 10 full energy+force evaluations with the calculator cache cleared each time.

| Metric | Value |
|---|---|
| min | 92.1 ms/eval |
| median | 94.0 ms/eval |
| max | 1196.3 ms/eval (first-call outlier) |
| energy | -49242.456415 eV |

The engine's **183.78 ms/eval** is **1.95x slower** than the PyTorch reference for the full model. The engine's backbone-only ablation is **87.24 ms**, close to the full PyTorch time, so the gap is dominated by the polar kernels rather than by the nonlinear backbone. This is the ratio that the subsequent kernel work has to close; the v28-to-v29 speedup alone does not establish an engine-vs-PyTorch advantage.

Caveat: one frame, one process, and the max includes a first-call outlier, so the median (not the mean) is the comparison quantity.

## 3.8 Figures

The reading package adds the higher-level diagnostic figures (relative paths from this `docs/` directory):

See reading guide for the reading order, data provenance, and reproduction entry points.

## 3.9 Kernel-level decomposition

Source: `docs/design/KERNEL_PROFILE_2026-10-19.md` and the CUPTI/Kokkos traces under `mlip/bench/kernel_profile_2026-10-19/`. Hardware counters were blocked by the container/driver, so durations and launch shapes are measured; occupancy is a static estimate.

| Family | GPU time (ms) | % of 183.5 ms eval | Launches |
|---|---|---|---|
| polar pipeline (SCF/Coulomb/LE/UVU/dense) | 73.36 | 40.0% | 238 |
| e3nn scalar linear forward + reverse | 66.68 | 36.3% | 302 |
| backbone nonlinear (gates/gather/normalization) | 19.03 | 10.4% | 46 |
| tensor product (MH-1 fused paths) | 10.91 | 5.9% | 18 |
| MLP / dense | 3.44 | 1.9% | 65 |
| cuBLAS / external | 1.40 | 0.8% | 51 |

The eval is GPU-busy 96% of wall time, but **SM-weighted utilization is only 52%** (95.7/183.5 ms). The dominant defect is launch shape, not launch count: 39 kernels run on **1x1x1 grids** (128 threads) and consume **47.8 ms** while using 0.042 ms of SM-weighted time. The two worst are `polar reverse dense` (25.74 ms) and `reverse nonlinear normalization` (12.74 ms), each one block. The largest single family is the `e3nn` scalar linear path (66.68 ms); its forward and reverse kernels are load/latency-bound and re-read a small weight matrix per sample. These are the targets of the in-flight kernel round.

# 4. Findings

## 4.1 Cost structure: backbone plus polar non-k kernels

Completed measurements give a consistent picture:

- **128 atoms, same model:** backbone-only 87.24 ms -> full 183.50 ms. The polar layer adds 96.26 ms (52.5% of the full eval, 2.10x).
- **432 atoms, N2/33 A, same model:** full 475.883 ms/step vs polar-off 190.738 ms/step. The polar layer adds **285.145 ms (59.9%)**; the backbone accounts for 190.738 ms (40.1%).
- **Stage profile at 128 atoms** (`polar_profile_8eval_v29.log`, representative uncontended eval): backbone + stage 1 ~36.6 ms; polar forward pipeline ~20 ms, of which the k-space Coulomb stage is ~14.0 ms; reverse/replay and node forces ~53 ms. The k-space slice is therefore only ~7.6% of that eval, while the nonlinear MLP/UVU/replay kernels dominate the polar cost.

**Long-range share.** The full polar layer is ~60% of the 432-atom step in the completed coarse ablation. The finer figure that the ongoing profiling round attributes to the k-space/long-range slice, **~20% of the 432-atom step**, is an in-flight estimate: the dedicated 432-atom stage profile (`polar_profile_n2_33a.py`) had produced no output at the evidence cut-off. Treat the ~20% as provisional until that JSON/report exists; the 60% full-polar ablation and the 40% backbone share are directly measured.

**Bottleneck:** the nonlinear backbone (roughly 40-48%) plus the polar non-k kernels (UVU/MLP reverse, replay, local-electron, node forces). Optimizing only the k-space Ewald/RBE stage cannot remove most of the cost.

## 4.2 MH-1 is architecturally blocked

The rejection is a family-shape check, not a missing export or a flag. The model's second interaction differs on edge channels, hidden irreps, product output, and readout class; the checkpoint matches the JSON exactly. Route A (re-export) is infeasible without retraining/distillation, and Route C (relaxing only the gate) would run unprepared conditioned-MLP tables and is numerically wrong. The staged route estimates are engineering estimates, not measurements:

| Route | Scope | Estimated speedup | Effort |
|---|---|---|---|
| B1 | family analysis + pair conditioning + generic streamed, FP32 enablement | ~0.5-1.0x (not a speed guarantee) | 5-8 days |
| B2 | B1 + v4 contract + JIT host/device artifacts, polar materialized | target 1.5-3x backbone, ~1.2-1.5x full model | +15-25 days |
| B3 | full polar direct/JIT | up to ~1.5-2x full model | +10-20 days |
| B4 | LAMMPS direct/JIT wiring | required to change the measured LAMMPS timings | +10-15+ days |

None of these estimates is included in the measured tables above.

## 4.3 RBE and 1-hop are rejected by the force gate

Both are genuine performance ideas, and both fail the minimum-accuracy rule:

- RBE: excellent statistical behavior (slope -0.49885, unbiased, bitwise same-seed reproducibility) but RMS_F 1e-2 to 4e-2 eV/A, i.e. 100-400x the 1e-4 eV/A force gate.
- 1-hop ghost: 2.86x faster than exact 2-hop, but dE 7.94e-5 eV (1.6x over the 5e-5 eV tolerance) and max dF 4.116e-3 eV/A (41x over the 1e-4 eV/A gate).
- Exact 2-hop is the only MPI mode that passes, and it is 28.1x slower than 1 rank on one GPU.

## 4.4 One GPU means MPI does not speed up

All MPI timings above were taken on one shared A100. The exact 2-hop ghost graph has ~20x more edges than the 1-rank graph in the 14.7 A box, and the 2-hop workspace exceeds a single card at 4 ranks (4.5-6.9 GiB per block). The upstream linear H1 message passing is also 3.38x slower at 2 ranks on the same GPU. MPI is a functional/correctness milestone, not a speedup, until per-layer feature exchange and multiple GPUs are available.

# 5. In-flight optimization round (no results claimed)

Other agents are actively optimizing the polar and backbone kernels. The following workstreams were observed on disk at the evidence cut-off. None is included in any number above.

| Workstream | Evidence | Current state | Reporting |
|---|---|---|---|
| Polar kernel optimization | rebuild logs symmetrix_rebuild_v29alloc*.log, ..._v29lealloc.log, ..._v29tune.log; no new timing JSON | in flight | pending |
| Backbone kernel optimization | symmetrix_rebuild_v31bb1.log (08:18) | in flight | pending |
| ncu / kernel profile | bench/kernel_profile_2026-10-19/; runs/full_v29/cupti_kernels.csv has header only | in flight | pending |
| MPI H1 message passing | logs/p5_lammps_build_v31h1.log; LAMMPS built successfully at 07:49, but no parity or timing JSON | in flight; build success is not a result | pending |
| 432-atom stage profile | bench/polar_profile_n2_33a.out is 0 bytes | in flight | pending |
| 33 A RBE force-error measurement | bench/polar_rbe_force_error_33a.py (08:15), no output JSON | in flight | pending |
| PyTorch reference speed | bench/polar_torch_reference_speed.py (08:14), no log | in flight | pending |
| leoff timing | model file exists; no timing JSON | not yet run | pending |

No engine source, model file, or agent bench output was modified while preparing this report.

# 6. Reproduction commands

All commands run from the MLIP bench workspace, which is the source of the gates and logs:

```
export MLIP=/workspace/group/shared/bza/task3_sampling_database/mlip
cd "$MLIP"

export CUDA_VISIBLE_DEVICES=0
export SYMMETRIX_BACKEND=cuda12-sm80
export SYMMETRIX_BACKEND_DESCRIPTOR_PATH="$MLIP/models/mace-polar/backend_cuda12_sm80.json"
export
```

```
LD_LIBRARY_PATH=/opt/matos-tools/cuda/12.9/lib64:/opt/matos-tools/conda/envs/matos-compute/lib
PY=/workspace/caches/conda/envs/polar/bin/python

# Accuracy gates
$PY bench/polar_module5_gate.py
$PY bench/polar_module6_gate.py
$PY bench/polar_module7_gate.py
$PY bench/polar_module8_gate.py
$PY bench/polar_endtoend_parity.py
$PY bench/polar_rbe_gate.py

# LAMMPS pair style and MD (requires the patched LAMMPS build)
$PY bench/polar_lammps_parity.py
$PY bench/polar_lammps_md.py

# Speed
$PY bench/polar_perf_report.py
$PY bench/polar_vs_linear_speed.py
$PY bench/polar_backbone_lammps.py
$PY bench/linear_streamed_compare.py
$PY bench/linear_mpi_speed.py
$PY bench/polar_homog_timing.py
$PY bench/polar_rbe_md_timing.py

# Pending / in-flight (no completed artifact at the cut-off)
$PY bench/polar_rbe_force_error_33a.py
$PY bench/polar_profile_n2_33a.py
$PY bench/polar_torch_reference_speed.py

Notes: - Do not run these on a shared GPU without exclusive access; the v29 absolute times are
contention-sensitive. - The patched LAMMPS binary used by the gates is
/workspace/group/shared/bza/external-repos/symmetrix-xl/symmetrix/build-lammps-aacd036df381800
a/lmp. - The engine patch is patches/symmetrix-xl-polar-support.patch (325,145 bytes).
```

# 7. Artifact index

## 7.1 In this repository (links are relative to `docs/`)

| Artifact | Path |
|---|---|
| v29 performance JSON | ../results/polar_perf_v29.json |
| v28 vs v29 figure | ../results/polar_perf_v28_vs_v29.png |
| Engine / model ablation JSON | ../results/engine_compare.json |
| Engine comparison figure | ../results/polar_engine_compare.png |
| Streamed vs materialized JSON | ../results/streamed_compare.json |
| MPI 1-rank JSON | ../results/polar_time_mpi1.json |
| MPI exact 2-hop JSON | ../results/polar_time_mpi2.json |
| MPI 1-hop JSON | ../results/polar_time_mpi2_onehop.json |
| Linear H1 MPI JSON | ../results/linear_mpi.json |
| MPI status figure | ../results/polar_mpi_status.png |
| LAMMPS parity, 1 rank | ../results/polar_lammps_parity.json |
| LAMMPS parity, 2 rank | ../results/polar_lammps_parity_mpi2.json |
| LAMMPS parity figure | ../results/polar_lammps_parity.png |
| LAMMPS NVT MD JSON | ../results/polar_md.json |
| LAMMPS NVT MD figure | ../results/polar_lammps_md.png |
| RBE gate JSON | ../results/polar_rbe_gate_v29.json |
| RBE gate figure | ../results/polar_rbe_gate_v29.png |
| MH-1 feasibility analysis | design/MH1_FASTPATH_FEASIBILITY.md |
| Key numeric rows (CSV) | ../results/comparison_summary.csv |

## 7.2 External source artifacts (workspace paths)

| Artifact | Path |
|---|---|
| v28 benchmark JSON | /workspace/group/shared/bza/task3_sampling_database/mlip/bench/polar_perf_v28.json |

| Artifact | Path |
|---|---|
| v29 benchmark JSON (copy) | /workspace/group/shared/bza/task3_sampling_database/mlip/bench/polar_perf.json |
| Engine comparison JSON | .../bench/lammps_parity/engine_compare/engine_compare.json |
| Homogeneous timings | .../bench/lammps_parity/homog_timing/homog_timing.json |
| RBE 33 A timings | .../bench/lammps_parity/rbe_md_timing/rbe_md_timing.json |
| Backbone ablation (N2) | .../bench/lammps_parity/backbone_timing/result.n2_full.json, result.n2_polaroff.json |
| Gate logs | .../mlip/logs/polar_module{5,6,7,8}_gate_v29*.log |
| End-to-end log | .../mlip/logs/polar_endtoend_v29final.log |
| RBE gate log | .../mlip/logs/polar_rbe_gate_v29opt.log |
| LAMMPS parity/MD logs | .../mlip/logs/polar_lammps_parity_v29final.log, polar_lammps_md_v29final.log, polar_lammps_md_mpi2_n32.log |
| Stage profile log | .../mlip/logs/polar_profile_8eval_v29.log |
| Engine patch | patches/symmetrix-xl-polar-support.patch |

# 8. Data-quality notes and missing artifacts

These are deliberate caveats, not inferred results:

1. **`results/polar_perf_v28.json` is absent from this repository.** The v28 benchmark JSON exists only at `mlip/bench/polar_perf_v28.json`; the v28 numbers in Section 3.1 come from there.
2. **v28 deterministic row:** the report's 6951.7 ms is an energy-only median; the energy+force benchmark JSON is 6966.79 ms. Section 3.1 uses the JSON.
3. **End-to-end relative error:** the v29 final log shows 3.2e-12 to 1.1e-11 across the 32 frames. The "<=4.2e-12" line in `REPORT_2026-10-19.md` matches only the CO2 subset; the tables above use the full 32-frame range.
4. **`results/polar_md.json` vs final log:** the JSON records 599.584 ms/step (29.9792 s loop); the final v29 log records **493.662 ms/step** (24.6831 s loop) with identical thermo. The difference is GPU contention; Section 2.3 uses the final log and notes the JSON value here.
5. **Linear generic timing spread:** the engine-compare run measured 7.219 ms with `auto -> generic`; the explicit streamed compare measured 19.291 ms on the same path. Both are reported; materialized (6.6102 ms) is the stable linear reference.
6. **`n2_homog_pilot.log` is empty (0 bytes).** That path was a stale launcher log; the completed 1 ps pilot is `runs/n2-homog-pilot/run-status.json` (2000 production steps, 537.99 ms/step; see Section 3.5).
7. **432-atom long-range share:** the ~20% k-space estimate is in flight; the completed coarse ablation gives 59.9% for the *whole polar layer*. The dedicated profile output was empty at the cut-off.
8. **33 A RBE force-error JSON, leoff timing, and MPI H1 validation/timing are still pending.** The PyTorch reference speed log now exists and is reported in Section 3.7; the ncu profile report now exists (`docs/design/KERNEL_PROFILE_2026-10-19.md`) but its optimization results remain in flight. No numbers were invented for pending items.
9. **4-rank OOM** is reported from the v30 report; no separate JSON artifact was produced.
10. Absolute timings are contention-sensitive; ratios from the same run are the reliable quantity.

*Prepared from completed project artifacts only. Rows added after the 08:20 UTC cut-off (PyTorch reference speed, 1 ps N2 pilot) cite their source logs; no numbers were estimated from in-flight work, and no other agents' bench outputs were modified.*

# Figure gallery

The key diagnostic figures from docs/figures/. Full captions are in the report body and in the reading guide.

**Figure 1. Where the time goes: backbone vs long-range vs polar kernels.**

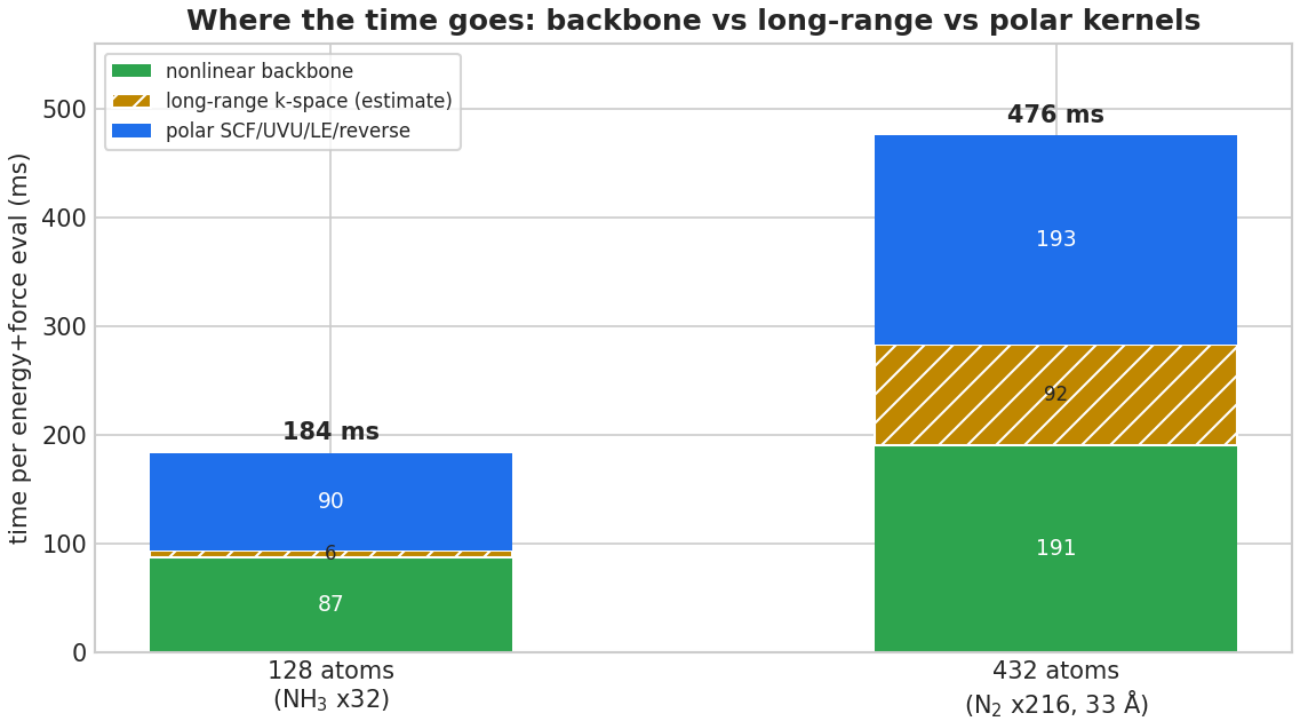



**Figure 2. v28 vs v29: P5 LAMMPS wiring and the 37.8x single-rank speedup.**

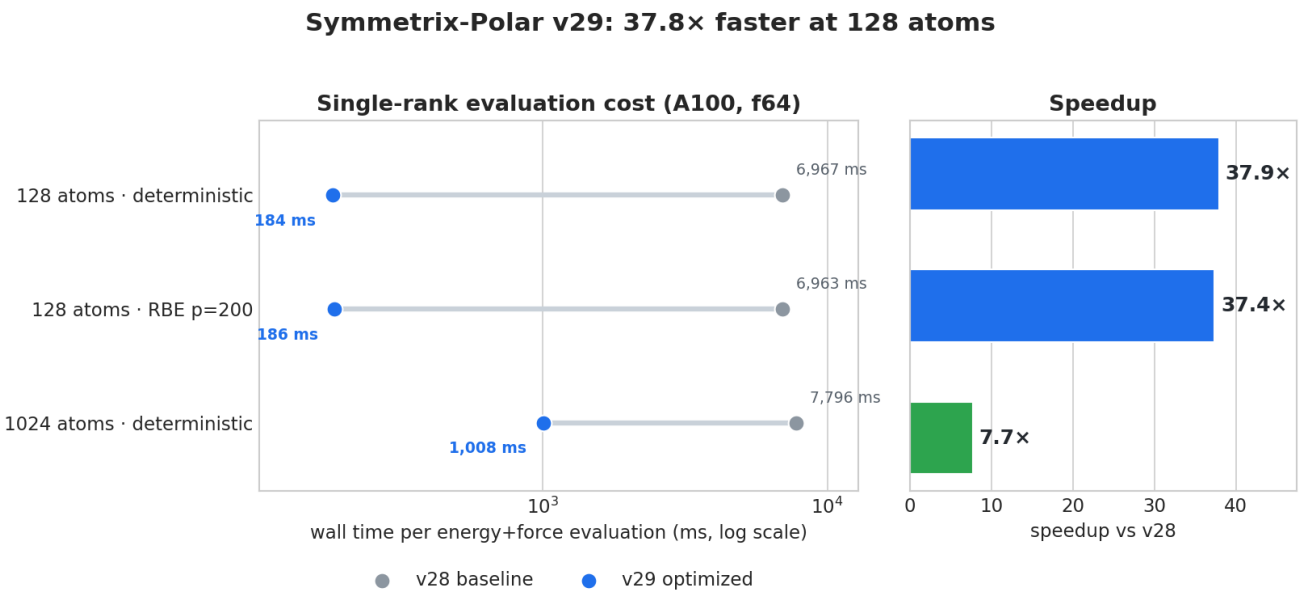


**Figure 3. RBE is faster but fails the 1e-4 eV/Å force gate.**

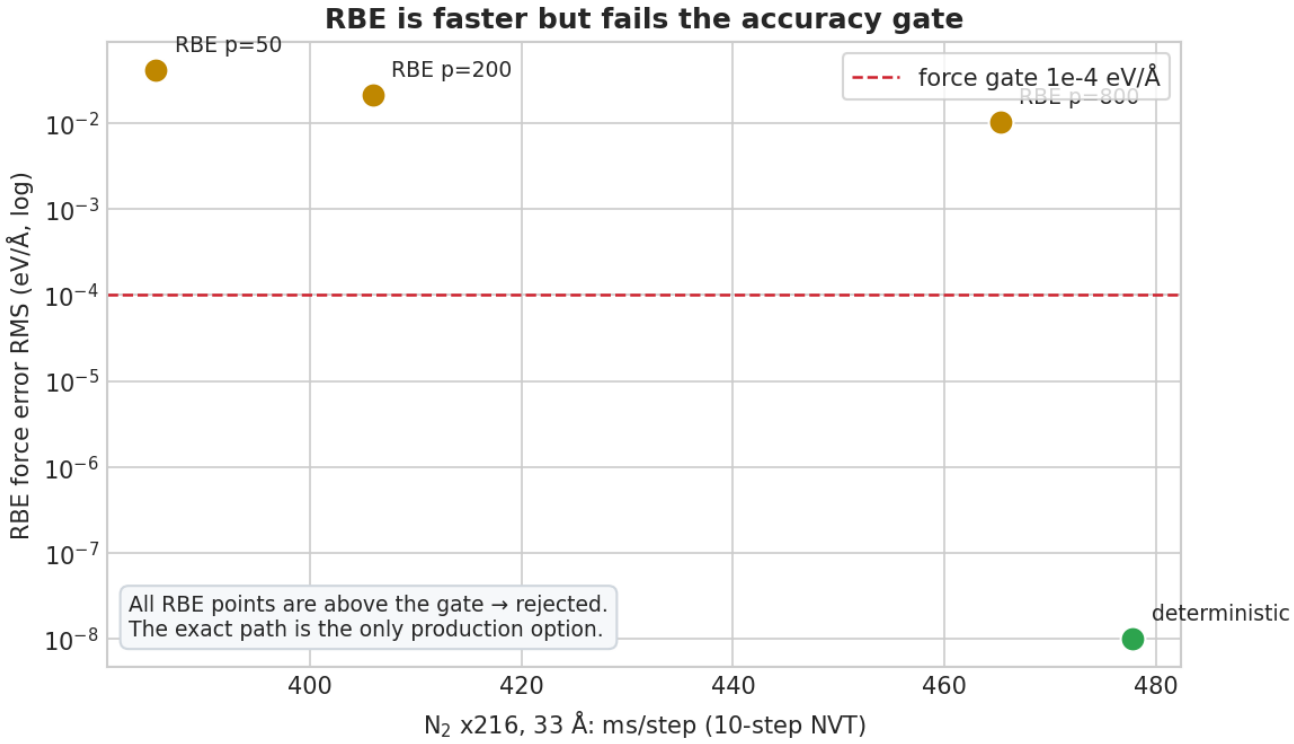



**Figure 4. RBE force error vs batch size against the hard force gate.**

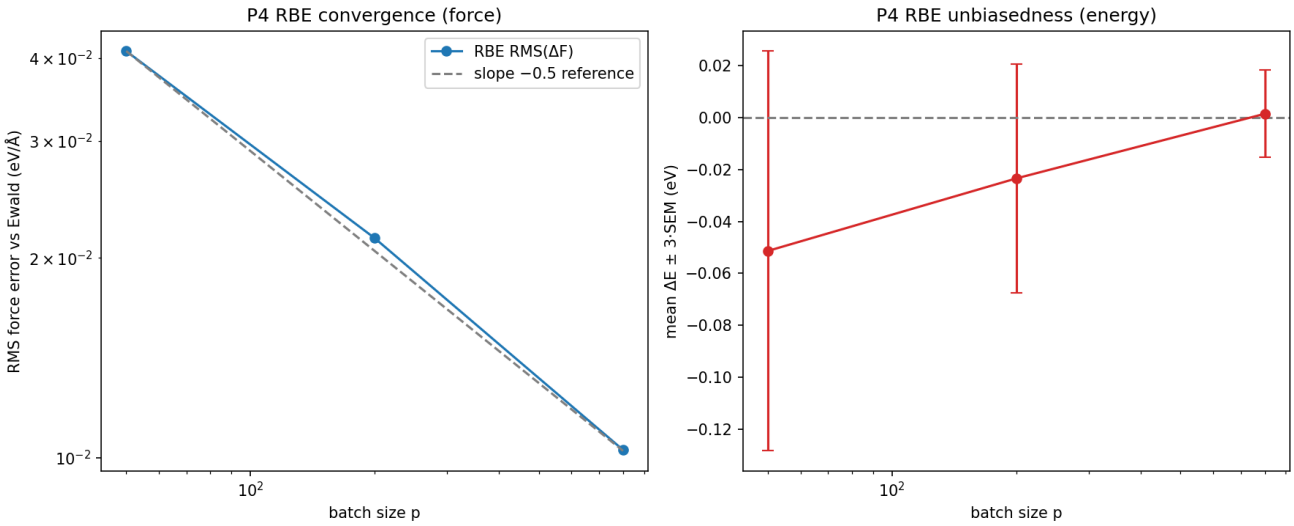


**Figure 5. All accuracy gates pass with large margin.**

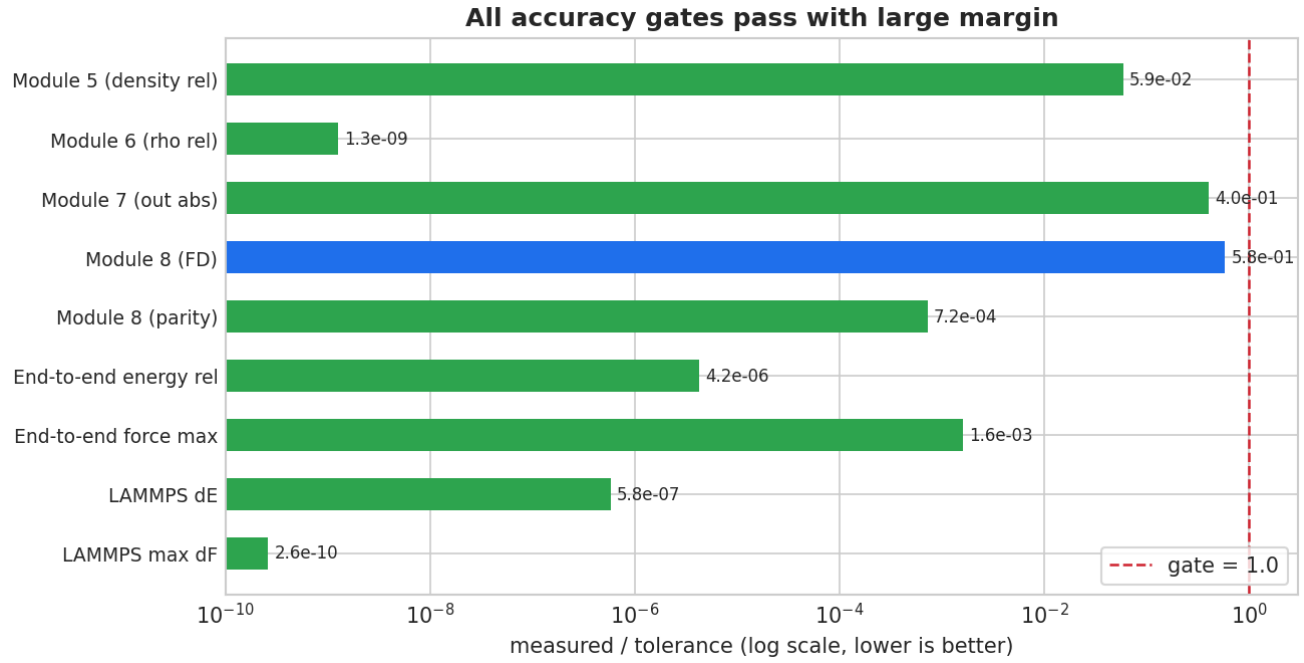



**Figure 6. Homogeneous MD per-step cost and PNAS-length 1 ns extrapolation.**

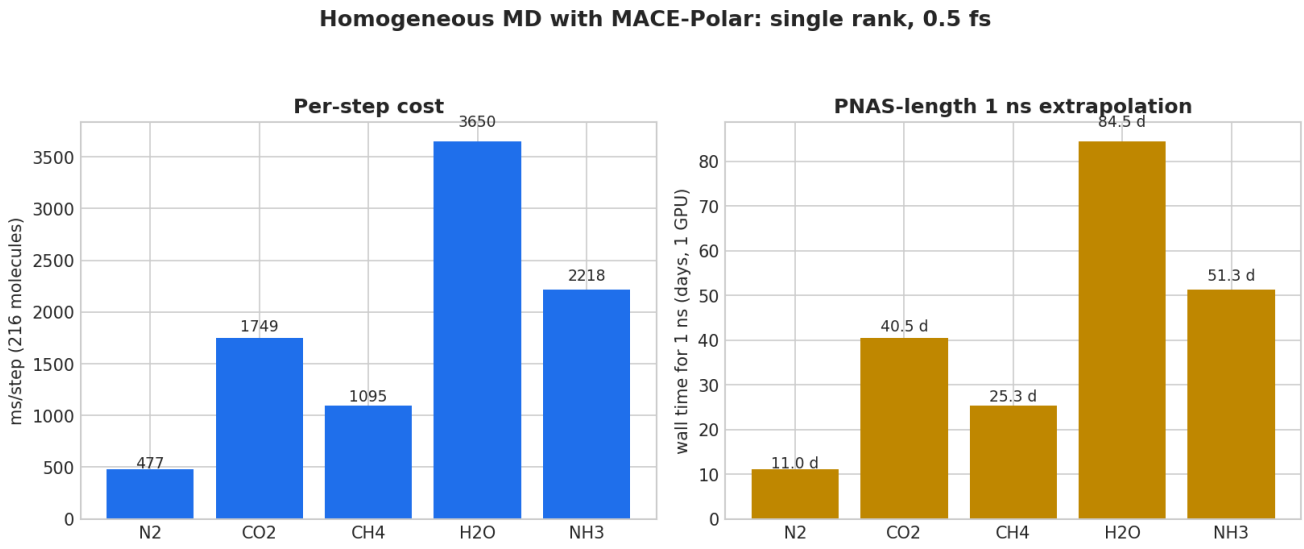


**Figure 7. Engine/model cost: linear MACE vs nonlinear backbone vs full PolarMACE.**

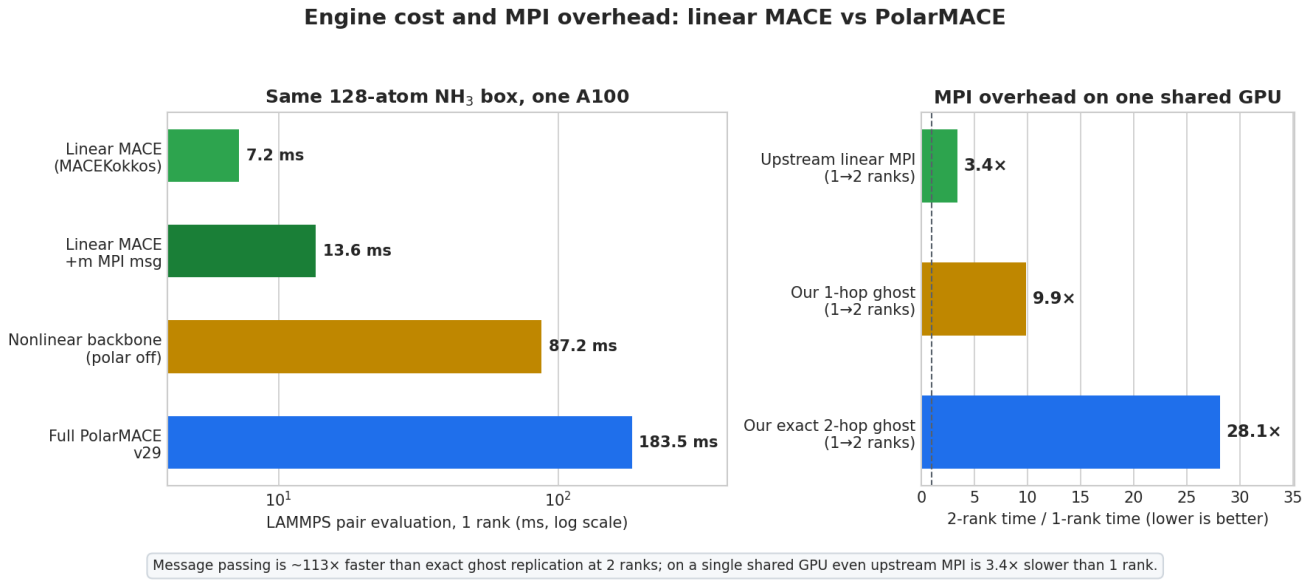



**Figure 8. MPI on one shared A100: correctness achieved, no speedup yet.**

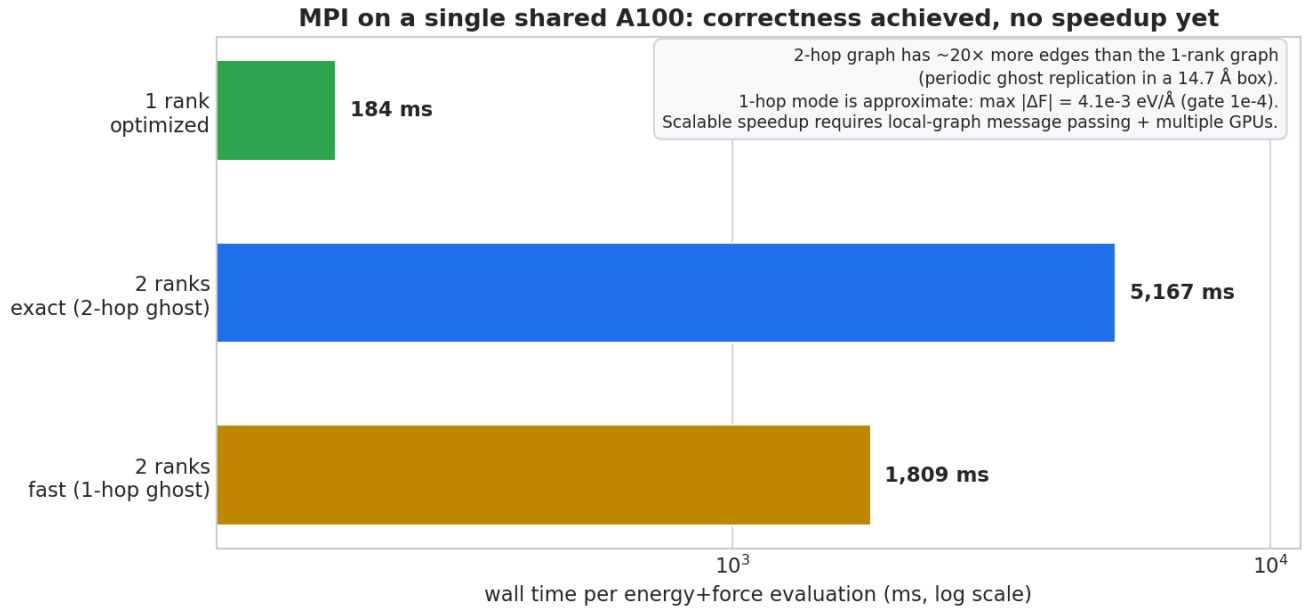


**Figure 9. LAMMPS single-point force parity and residual distribution.**

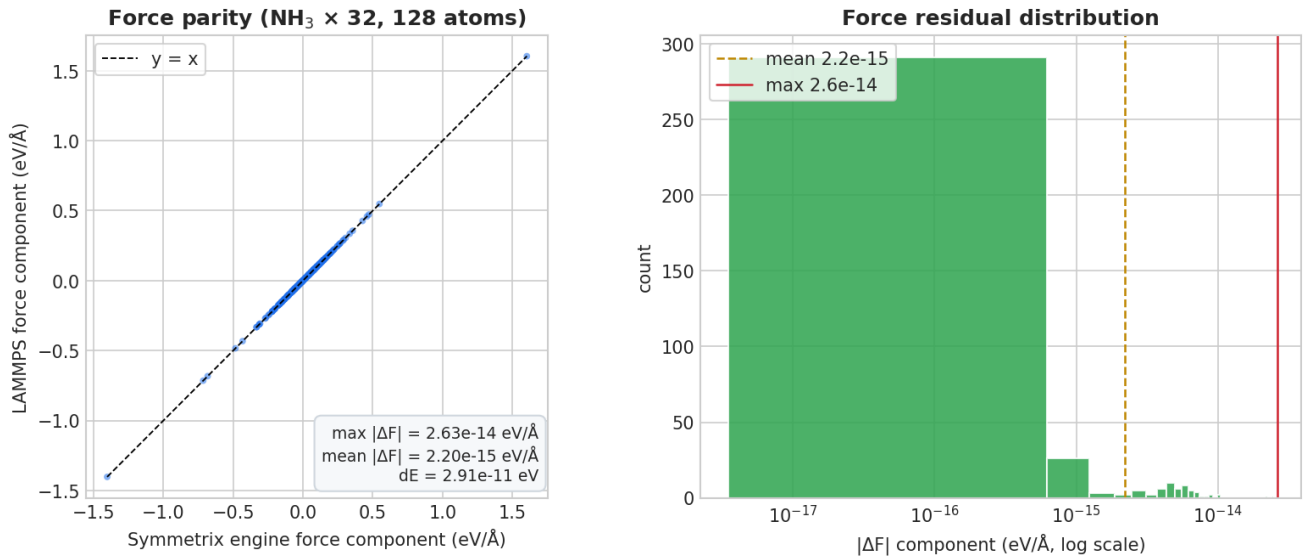



**Figure 10. NVT MD temperature and energy stability.**

